

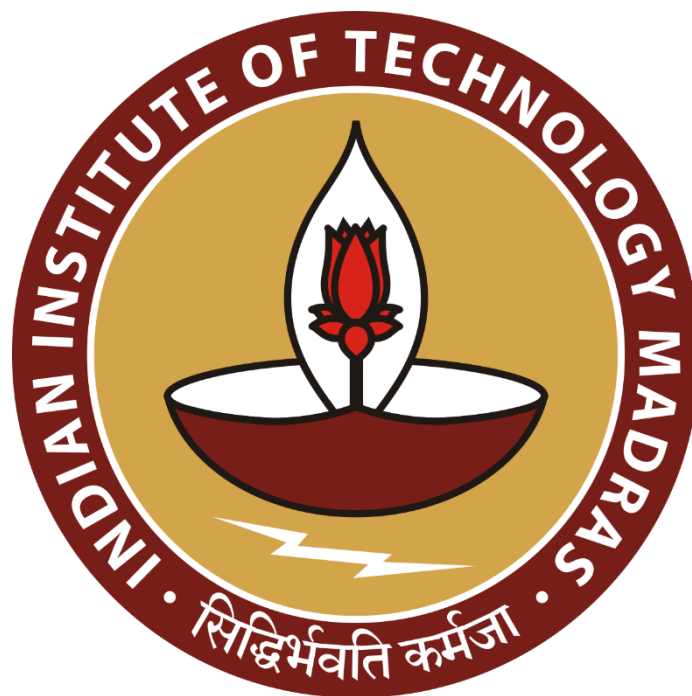
“Optimizing Inventory Management and Profitability for Sudha Milk Parlor”

A Proposal Report for the BDM Capstone Project

Submitted by

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Declaration Statement

I am working on a Project titled “**Optimizing Inventory Management and Profitability for Sudha Milk Parlor**”. I extend my appreciation to **Sudha Milk Parlor**, for providing the necessary resources that enabled me to conduct my project.

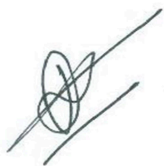
I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.



Signature of Candidate

Name: **Rahul Ramu**

Date: **30.01.2025**

1 Executive Summary and Title

“Optimizing Inventory Management and Profitability for Sudha Milk Parlor”

Sudha Milk Parlor is a state co-operative dairy shop, located in Hajipur, Vaishali, Bihar. The shop follows a B2C (Business-to-Consumer) business model, directly selling dairy related products to customers.

One of the biggest challenges faced by the business is poor inventory management. The identification of goods that are not purchased often by customers is lacking in the business, thus leading to overstocking of unsold products. Another major problem is the short shelf life of dairy products compared to other products. If any product is unsold over a longer period of time, then there are high chances of the product being expired.

These issues will be addressed by studying the purchasing behavior of customers based on the sales data of the business over a period of month. Through various data visualization , products that are not sold in large numbers will be identified and possibly replaced by best-selling products. Additionally, in order to make sure that none of the products expire before being sold, a dynamic pricing strategy can be implemented to reduce the price of such products towards the expiration date. This will ensure that there is at least a minimal revenue instead of none at all. These implementations can assist the business in the efficient utilization of the inventory space and enhancing profitability.

2 Organization Background

Sudha Milk Parlor, located in Hajipur, Vaishali, Bihar, is a state co-operative dairy shop that has been serving customers for the last 19 years. The shop functions on every day of the week (including weekends) from 10 AM to 8 PM, closing only on major holidays such as Diwali and Holi. The average age of customers that regularly visit the shop lies between 35 and 40, prompting the business to orient its business strategy according to the needs and requirements of middle-aged consumers. The shop is located in neither a remote nor a prime location, leading to a moderate level of customer footfall. Even though it is a small-scale business (in terms of infrastructure and operations), it has been able to generate an appreciable amount of profit during its period of operation.

3 Problem Statement

3.1 Inventory Mismanagement

The business currently lacks a strategy to identify slow-moving products, leading to inefficient utilization of the limited inventory space and ultimately affecting its overall profitability.

3.2 Product Wastage

The business does not have a clearance strategy for products that are nearing expiration, causing financial losses and an increased amount of wastage.

4 Background of the Problem

Inventory Mismanagement

Given the small-scale nature of the shop, it is of utmost importance for it to efficiently manage the limited inventory space. One of the major causes of this problem is the lack of data-dependent decision-making strategies. The shop does not focus on analyzing profits through individual products that are being sold, but only considers the overall profit over a period of time. This leads to overstocking of products that are less-preferred by customers, which not only occupies the limited inventory space, but also directly affects the shop's profitability.

Product Wastage

Due to the short shelf life nature of the products that are being sold in the shop, proper measures need to be taken to ensure that no products are wasted after being unsold past their expiration date. The main reason for the wastage of products is the fixed pricing strategy of the shop. Whether products are fresh or nearing expiration, they are sold at the same price, which discourages the sales of such products. Hence, the shop is forced to dispose of these products, incurring losses that could be partially, if not entirely, avoided.

5 Problem-Solving Approach

Data Collection

The first step in solving the identified problem statements is to collect relevant data from the shop over a period of time (in this case, 30 days). This data will provide information about two major aspects:

- (i) A list of different products that the shop purchases from suppliers, along with their quantities and cost prices.
- (ii) A list of different products sold every day in the shop, along with their quantities and selling prices.

This data is crucial for understanding customer preferences and analyzing the profit trends of individual products.

Approach towards solving the problem statements

(I) Profit Analysis

The profitability of individual products will be assessed using the data collected from the shop, by comparing the total amount of money spent on each of the products (i.e.) the total investment in each product and their corresponding sales revenue. This analysis of profitability will be further simplified using several data visualization techniques, such as bar graphs, to highlight trends. Depending on the results from this analysis, it will be possible to distinguish slow-moving products from high-demand products. These products can then be purchased in smaller quantities in the future rather than the actual pre-planned quantities. This analysis will not only help in identifying slow-moving products and help resolve the issue of inventory mismanagement, but it will also aid in identifying products that are sold in good numbers. This is really crucial to enhance the business's profits by purchasing excess quantities of such products instead of the slow-moving ones.

(II) Wastage Analysis

In continuation with the profit analysis, it will be easy to identify products that are unsold over extended periods of time and are nearing their expiration date. Once such products are identified, a dynamic pricing strategy will then be implemented and their selling prices will be drastically reduced (the initial revised selling price will be set at three-fourths of the actual selling price). As the expiration date approaches, the selling price can be reduced even further to increase the chances of selling such products. Such a pricing strategy is crucial to attract customers and avoid wastages, while also contributing some amount of revenue to the business's profits.

Hence, the profit analysis of individual products can help address not just one, but two crucial problems associated with the business. This data-dependent approach will enable the shop to make better purchasing decisions, reduce financial losses, and enhance the overall profitability.

6 Expected Timeline

6.1 Work Breakdown Structure

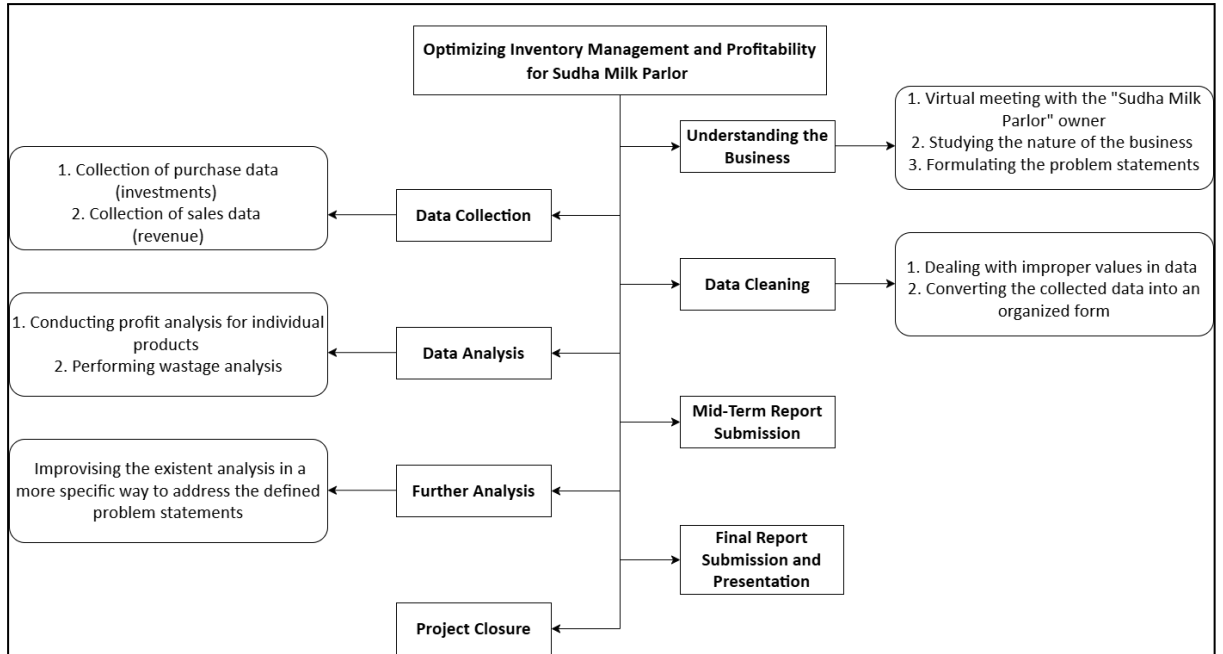


Figure 1: Work breakdown structure

6.2 Gantt chart

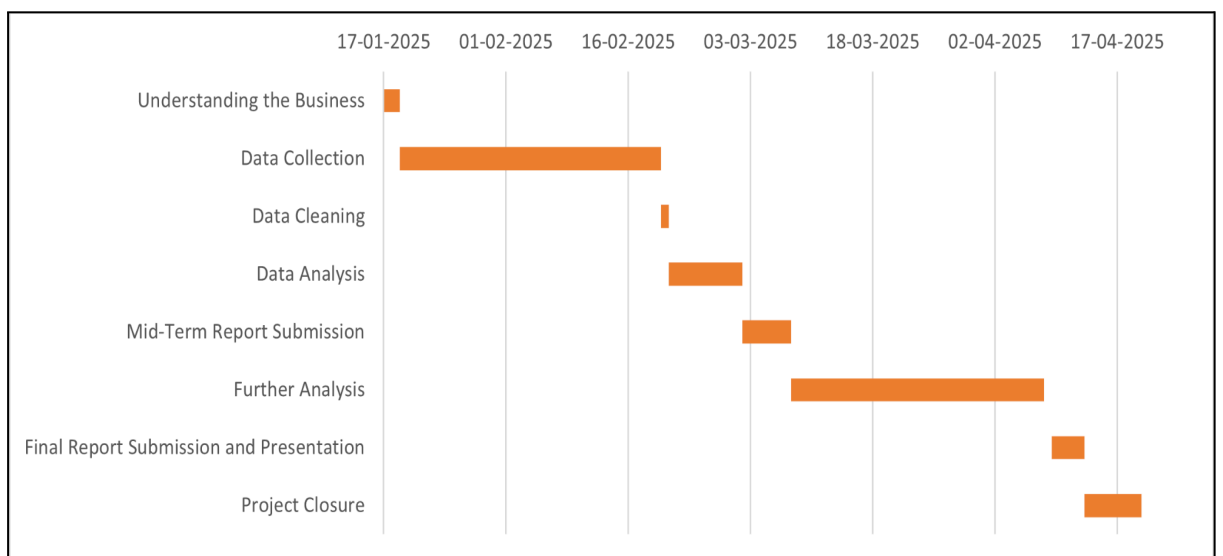


Figure 2: Gantt Chart

7 Expected Outcome

7.1 Optimal Utilization of Inventory Space

By reducing the frequent purchase of slow-moving products, the shop can utilize the inventory space in an efficient way by storing the high-demand products. This will also help in preventing the unnecessary overstocking of items.

7.2 Minimized Product Wastage

By implementing a dynamic pricing strategy for products nearing expiration, the shop can increase the probability of selling such products, thereby reducing excess product wastage and minimizing financial losses.

7.3 Increased Profitability

By limiting the purchase of slow-moving products and giving more priority to the fast-moving products, the shop can boost the chance of achieving higher profitability compared to the current profit levels.